

Chapter 4 Vector Spaces

Section 4.4: Coordinate Systems

Section 4.5: The Dimension of a Vector Space

Section 4.6: Change of Basis

The Unique Representation Theorem

- **Theorem:** Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for vector space V . Then for each $\mathbf{x}$ in V , there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$\mathbf{x} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n \quad (1)$$

- **Proof:** Since B spans V , there exist scalars such that (1) holds. Suppose $\mathbf{x}$ also has the representation for scalars $d_1, \dots, d_n$.

$$\mathbf{x} = d_1 \mathbf{b}_1 + \dots + d_n \mathbf{b}_n$$

The Unique Representation Theorem

- Then, subtracting, we have

$$0 = \mathbf{x} - \mathbf{x} = (c_1 - d_1)\mathbf{b}_1 + \dots + (c_n - d_n)\mathbf{b}_n \quad (2)$$

- Since B is linearly independent, the weights in (2) must all be zero. That is, $c_j = d_j$ for $1 \leq j \leq n$.
- **Definition:** Suppose $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for V and $\mathbf{x}$ is in V . **The coordinates of $\mathbf{x}$ relative to the basis B** (or the B -coordinate of $\mathbf{x}$) are the weights $c_1, \dots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n$$

The Unique Representation Theorem

- If $c_1, \dots, c_n$ are the **B**-coordinates of $\mathbf{x}$, then the vector in $\mathbb{R}^n$

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is the **coordinate vector of $\mathbf{x}$** (relative to B), or the **B-coordinate vector of $\mathbf{x}$** .

- The mapping $\mathbf{x} \mapsto [\mathbf{x}]_B$ is the coordinate mapping (determined by B).

Coordinates in $\mathbb{R}^n$

- When a basis B for $\mathbb{R}^n$ is fixed, the B -coordinate vector of a specified $\mathbf{x}$ is easily found, as in the example below.

- **Example:** Let $\mathbf{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, and

$B = \{\mathbf{b}_1, \mathbf{b}_2\}$. Find the coordinate vector $[\mathbf{x}]_B$ of $\mathbf{x}$ relative to B .

- **Solution:** The B -coordinate c_1, c_2 of $\mathbf{x}$ satisfy

$$c_1 \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{\mathbf{b}_1} + c_2 \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{\mathbf{b}_2} = \underbrace{\begin{bmatrix} 4 \\ 5 \end{bmatrix}}_{\mathbf{x}}$$

Coordinates in $\mathbb{R}^n$

or

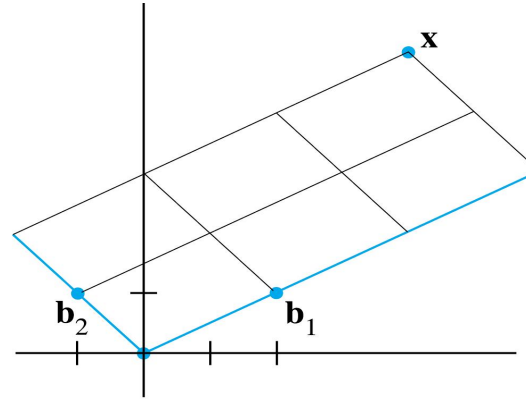
$$\begin{array}{cc} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} & (3) \\ \mathbf{b}_1 & \mathbf{b}_2 \quad \mathbf{x} \end{array}$$

- This equation can be solved by row operations on an augmented matrix or by using the inverse of the matrix on the left.
- In any case, the solution is $c_1 = 3$, $c_2 = 2$.
- Thus $\mathbf{x} = 3\mathbf{b}_1 + 2\mathbf{b}_2$ and

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Coordinates in $\mathbb{R}^n$

- See the following figure.



The $\mathcal{B}$ -coordinate vector of $\mathbf{x}$ is $(3, 2)$.

- The matrix in (3) changes the $\mathcal{B}$ -coordinates of a vector $\mathbf{x}$ into the standard coordinates for $\mathbf{x}$.
- An analogous change of coordinates can be carried out in $\mathbb{R}^n$ for a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$.
- Let $P_B = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_n]$

Coordinates in $\mathbb{R}^n$

- Then the vector equation

$$\mathbf{x} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 + \dots + c_n \mathbf{b}_n$$

is equivalent to

$$\underline{\mathbf{x} = P_B [\mathbf{x}]_B} \quad (4)$$

- P_B is called the **change-of-coordinates matrix** from B to the standard basis in $\mathbb{R}^n$.
- Left-multiplication by P_B transforms the coordinate vector $[\mathbf{x}]_B$ into $\mathbf{x}$.
- Since the columns of P_B form a basis for $\mathbb{R}^n$, P_B is invertible (by the Invertible Matrix Theorem).

Coordinates in $\mathbb{R}^n$

- Left-multiplication by P_B^{-1} converts $\mathbf{x}$ into its B-coordinate vector:

$$P_B^{-1}\mathbf{x} = [\mathbf{x}]_B$$

- The correspondence $\mathbf{x} \mapsto [\mathbf{x}]_B$, produced by P_B^{-1} , is the coordinate mapping.
- Since P_B^{-1} is an invertible matrix, the coordinate mapping is a one-to-one linear transformation from $\mathbb{R}^n$ onto $\mathbb{R}^n$, by the Invertible Matrix Theorem.

The Coordinate Mapping

- **Theorem:** Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V . Then the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_B$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

- **Proof:** Take two typical vectors in V , say,

$$\mathbf{u} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n$$

$$\mathbf{w} = d_1 \mathbf{b}_1 + \dots + d_n \mathbf{b}_n$$

- Then, using vector operations,

$$\mathbf{u} + \mathbf{w} = (c_1 + d_1) \mathbf{b}_1 + \dots + (c_n + d_n) \mathbf{b}_n$$

The Coordinate Mapping

- It follows that

$$[\mathbf{u} + \mathbf{w}]_{\mathbf{B}} = \begin{bmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = [\mathbf{u}]_{\mathbf{B}} + [\mathbf{w}]_{\mathbf{B}}$$

- So the coordinate mapping preserves addition.
- If r is any scalar, then

$$r\mathbf{u} = r(c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n) = (rc_1)\mathbf{b}_1 + \dots + (rc_n)\mathbf{b}_n$$

The Coordinate Mapping

- So

$$\begin{bmatrix} r\mathbf{u} \end{bmatrix}_{\mathbf{B}} = \begin{bmatrix} r c_1 \\ \vdots \\ r c_n \end{bmatrix} = r \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = r \begin{bmatrix} \mathbf{u} \end{bmatrix}_{\mathbf{B}}$$

- Thus the coordinate mapping also preserves scalar multiplication and hence is a linear transformation.
- The linearity of the coordinate mapping extends to linear combinations.
- If $\mathbf{u}_1, \dots, \mathbf{u}_p$ are in V and if $c_1, \dots, c_p$ are scalars, then

$$\begin{bmatrix} c_1 \mathbf{u}_1 + \dots + c_p \mathbf{u}_p \end{bmatrix}_{\mathbf{B}} = c_1 \begin{bmatrix} \mathbf{u}_1 \end{bmatrix}_{\mathbf{B}} + \dots + c_p \begin{bmatrix} \mathbf{u}_p \end{bmatrix}_{\mathbf{B}}$$

The Coordinate Mapping

- The coordinate mapping in Theorem 8 is an important example of an *isomorphism* from V onto $\mathbb{R}^n$.
- In general, a one-to-one linear transformation from a vector space V onto a vector space W is called an **isomorphism** from V onto W .
- *Every vector space calculation in V is accurately reproduced in W , and vice versa.*
- In particular, any real vector space with a basis of n vectors is indistinguishable from $\mathbb{R}^n$.

The Coordinate Mapping

• **Example:** Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$,

and $B = \{\mathbf{v}_1, \mathbf{v}_2\}$. Then B is a basis for $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Determine if $\mathbf{x}$ is in H , and if it is, find the coordinate vector of $\mathbf{x}$ relative to B .

The Coordinate Mapping

- **Solution:** If $\mathbf{x}$ is in H , then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

- The scalars c_1 and c_2 , if they exist, are the B-coordinates of $\mathbf{x}$.

The Coordinate Mapping

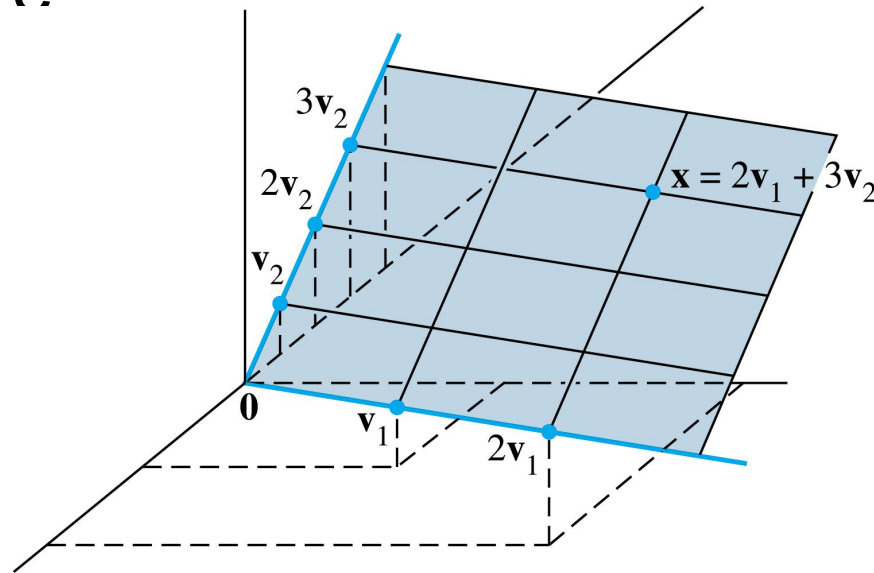
- Using row operations, we obtain

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

- Thus $c_1 = 2$, $c_2 = 3$ and $[\mathbf{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

The Coordinate Mapping

- The coordinate system on H determined by B is shown in the following figure



A coordinate system on a plane H in $\mathbb{R}^3$.

Section 4.4: Coordinate Systems

Section 4.5: The Dimension of a Vector Space

Section 4.6: Change of Basis

Dimension of a Vector

- **Theorem:** If a vector space V has a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.
- **Proof:** Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be a set in V with more than n vectors.

The coordinate vectors $[\mathbf{u}_1]_B, \dots, [\mathbf{u}_p]_B$ form a linearly dependent set in $\mathbb{R}^n$, because there are more vectors (p) than entries (n) in each vector.

Dimension of a Vector

- So there exist scalars $c_1, \dots, c_p$, not all zero, such that

$$c_1[u_1]_{\mathbb{B}} + \dots + c_p[u_p]_{\mathbb{B}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \text{The zero vector in } \mathbb{R}^n$$

Since the coordinate mapping is a linear transformation,

$$[c_1 u_1 + \dots + c_p u_p]_{\mathbb{B}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Dimension of a Vector

- The zero vector on the right displays the n weights needed to build the vector $c_1\mathbf{u}_1 + \dots + c_p\mathbf{u}_p$ from the basis vectors in B .
- That is, $c_1\mathbf{u}_1 + \dots + c_p\mathbf{u}_p = 0\mathbf{b}_1 + \dots + 0\mathbf{b}_n = \mathbf{0}$.
- Since the c_i are not all zero, $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is linearly dependent.
- The following theorem implies that if a vector space V has a basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, then each linearly independent set in V has no more than n vectors.

Dimension of a Vector

- **Theorem:** If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.
- **Proof:** Let B_1 be a basis of n vectors and B_2 be any other basis (of V).
- Since B_1 is a basis and B_2 is linearly independent, B_2 has no more than n vectors.
- Also, since B_2 is a basis and B_1 is linearly independent, B_2 has at least n vectors.
- Thus B_2 consists of exactly n vectors.

Dimension of a Vector

- **Definition:** If V is spanned by a finite set, then V is said to be **finite-dimensional**, and the **dimension** of V , written as $\dim V$, is the number of vectors in a basis for V . The dimension of the zero vector space $\{\mathbf{0}\}$ is defined to be zero. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.
- **Example:** Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a - 3b + 6c \\ 5a + 4d \\ b - 2c - d \\ 5d \end{bmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

Dimension of a Vector

- H is the set of all linear combinations of the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

- Clearly, $\mathbf{v}_1 \neq 0$, $\mathbf{v}_2$ is not a multiple of $\mathbf{v}_1$, but $\mathbf{v}_3$ is a multiple of $\mathbf{v}_2$.
- By the Spanning Set Theorem, we may discard $\mathbf{v}_3$ and still have a set that spans H .

Subspaces of a Finite-Dimensional Space

- Finally, $\mathbf{v}_4$ is not a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.
- So $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is linearly independent and hence is a basis for H .
- Thus $\dim$

$$H = 3$$

Subspaces of a Finite-Dimensional Space

Example: The subspaces of $\mathbb{R}^3$ can be classified by dimension.

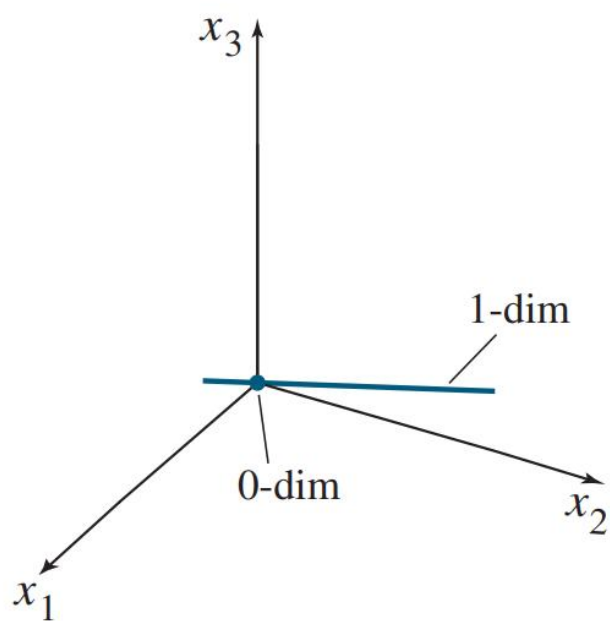
0-dimensional subspaces. Only the zero subspace.

1-dimensional subspaces. Any subspace spanned by a single nonzero vector. Such subspaces are lines through the origin.

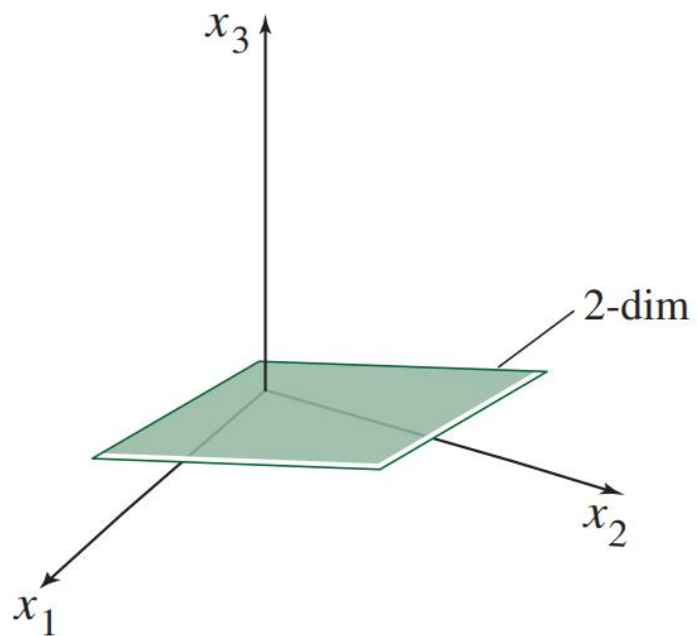
2-dimensional subspaces. Any subspace spanned by two linearly independent vectors. Such subspaces are planes through the origin.

3-dimensional subspaces. Only $\mathbb{R}^3$ itself. Any three linearly independent vectors in $\mathbb{R}^3$ span all of $\mathbb{R}^3$, by the Invertible Matrix Theorem. ■

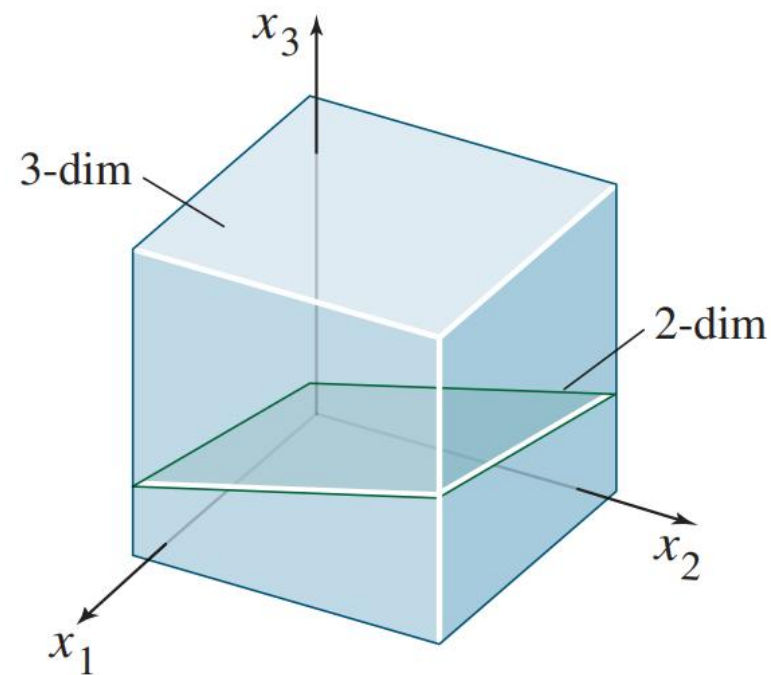
Subspaces of a Finite-Dimensional Space



(a)



(b)



(c)

Subspaces of a Finite-Dimensional Space

- **Theorem:** Let H be a subspace of a finite-dimensional vector space V . Any linearly independent set in H can be expanded, if necessary, to a basis for H . Also, H is finite-dimensional and

$$\dim H \leq \dim V$$

Subspaces of a Finite-Dimensional Space

- **Proof:** If $H = \{\mathbf{0}\}$, then certainly $\dim H = 0 \leq \dim V$.
- Otherwise, let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be any linearly independent set in H .
- If S spans H , then S is a basis for H .
- Otherwise, there is some $\mathbf{u}_{k+1}$ in H that is not in $\text{Span } S$.

Subspaces of a Finite-Dimensional Space

- But then $\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}\}$ will be linearly independent, because no vector in the set can be a linear combination of vectors that precede it.
- So long as the new set does not span H , we can continue this process of expanding S to a larger linearly independent set in H .
- But the number of vectors in a linearly independent expansion of S can never exceed the dimension of V .

So eventually the expansion of S will span H and hence will be a basis for H , and $\dim H \leq \dim V$.

The Basis Theorem

- **Theorem:** Let V be a p -dimensional vector space, $p \geq 1$.
 - a. Any linearly independent set of exactly p elements in V is automatically a basis for V .
 - b. Any set of exactly p elements that spans V is automatically a basis for V .

The Basis Theorem

- **Proof:** By Theorem, a linearly independent set S of p elements can be extended to a basis for V .
- But that basis must contain exactly p elements, since $\dim V = p$.
- So S must already be a basis for V .
- Now suppose that S has p elements and spans V .

The Basis Theorem

- Since V is nonzero, the Spanning Set Theorem implies that a subset S' of S is a basis of V .
- Since $\dim V = p$, S' must contain p vectors.
- Hence $S = S'$

The Rank Theorem

- **Definition:** The **rank** of A is the dimension of the column space of A .
- Since Row A is the same as Col A^T , the dimension of the row space of A is the rank of A^T .
- The dimension of the null space is called the **nullity** of A .

The Rank Theorem

- **Theorem:** The dimensions of the column space and the row space of an $m \times n$ matrix A are equal.
- This common dimension, the rank of A , also equals the number of pivot positions in A and satisfies the equation

$$\text{rank } A + \text{nullity } A = n$$

The Rank Theorem

- **Proof:** By Theorem, rank A is the number of pivot columns in A .
- Equivalently, rank A is the number of pivot positions in an echelon form B of A .
- Since B has a nonzero row for each pivot, and since these rows form a basis for the row space of A , the rank of A is also the dimension of the row space.
- The nullity A equals the number of free variables in the equation $A\mathbf{x} = \mathbf{0}$.
- Expressed another way, the nullity A is the number of columns of A that are *not* pivot columns.

The Rank Theorem

- (It is the number of these columns, not the columns themselves, that is related to $\text{Nul } A$).
- Obviously,

$$\left\{ \begin{array}{c} \text{number of} \\ \text{pivot columns} \end{array} \right\} + \left\{ \begin{array}{c} \text{number of} \\ \text{nonpivot columns} \end{array} \right\} = \left\{ \begin{array}{c} \text{number of} \\ \text{columns} \end{array} \right\}$$

- This proves the theorem.

The Rank Theorem

- **Example:**
 - a. If A is a 7×9 matrix with a two-dimensional null space, what is the rank of A ?
 - b. Could a 6×9 matrix have a two-dimensional null space?
- **Solution:**
 - a. Since A has 9 columns, $(\text{rank } A) + 2 = 9$, and hence $\text{rank } A = 7$.

The Rank Theorem

b. No. If a 6×9 matrix, call it B , has a two-dimensional null space, it would have to have rank 7, by the Rank Theorem.

But the columns of B are vectors in $\mathbb{R}^6$, and so the dimension of $\text{Col } B$ cannot exceed 6; that is, $\text{rank } B$ cannot exceed 6.

Rank and Nullity

- **Example:** Find the rank and nullity of A .

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Rank and Nullity

- **Solution:** Row reduce the augmented matrix $[A \ 0]$ to echelon form:

$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are three free variable— x_2 , x_4 and x_5 .
- Hence the nullity $A = 3$.
- Also rank $A = 2$ because A has two pivot columns.

The Invertible Matrix Theorem

- **Theorem:** Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.
 - m. The columns of A form a basis of $\mathbb{R}^n$.
 - n. $\text{Col } A = \mathbb{R}^n$
 - o. $\text{Rank } A = n$
 - p. $\text{Nullity } A = 0$
 - q. $\text{Nul } A = \{\mathbf{0}\}$

The Invertible Matrix Theorem

- **Proof:** Statement (m) is logically equivalent to statements (e) and (h) regarding linear independence and spanning.
- The other five statements are linked to the earlier ones of the theorem by the following chain of almost trivial implications:

$$(g) \Rightarrow (n) \Rightarrow (o) \Rightarrow (p) \Rightarrow (q) \Rightarrow (d)$$

Section 4.4: Coordinate Systems

Section 4.5: The Dimension of a Vector Space

Section 4.6: Change of Basis

Change of Basis

- **Example:** Consider two bases $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$ for a vector space V , such that

$$\mathbf{b}_1 = 4\mathbf{c}_1 + \mathbf{c}_2 \quad \text{and} \quad \mathbf{b}_2 = -6\mathbf{c}_1 + \mathbf{c}_2 \quad (1)$$

- Suppose

$$\mathbf{x} = 3\mathbf{b}_1 + \mathbf{b}_2 \quad (2)$$

- That is, suppose $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Find $[\mathbf{x}]_{\mathcal{C}}$.

Change of Basis

- **Solution** Apply the coordinate mapping determined by $\mathcal{C}$ to $\mathbf{x}$ in (2). Since the coordinate mapping is a linear transformation,

$$\begin{aligned} [\mathbf{x}]_{\mathcal{C}} &= [3\mathbf{b}_1 + \mathbf{b}_2]_{\mathcal{C}} \\ &= [3\mathbf{b}_1]_{\mathcal{C}} + [\mathbf{b}_2]_{\mathcal{C}} \end{aligned}$$

- We can write the vector equation as a matrix equation, using the vectors in the linear combination as the columns of a matrix:

$$[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad (3)$$

Change of Basis

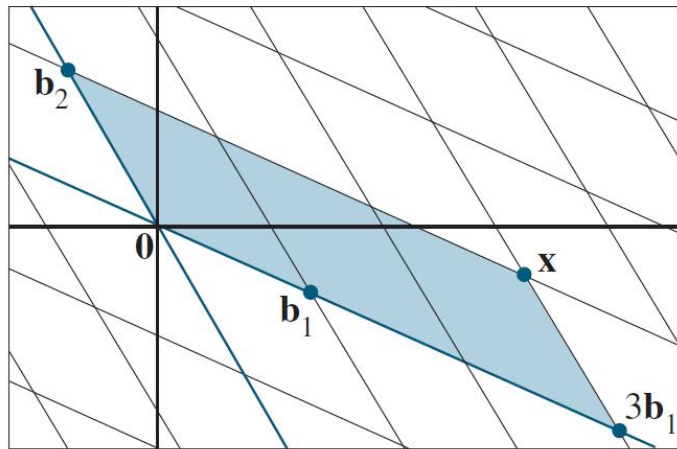
- This formula gives $[\mathbf{x}]_{\mathcal{C}}$, once we know the columns of the matrix.
From (1),

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \text{ and } [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$$

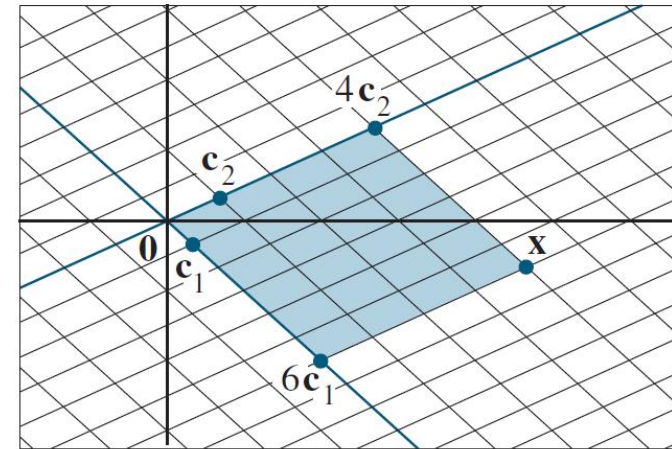
- Thus, (3) provides the solution:

$$[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Change of Basis



(a)



(b)

FIGURE 1 Two coordinate systems for the same vector space.

Change of Basis

- **Theorem:** Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_2\}$ for a vector space V . Then there is a unique $n \times n$ matrix ${}^P_{\mathcal{C}} \leftarrow \mathcal{B}$ such that

$$[\mathbf{x}]_{\mathcal{C}} = {}^P_{\mathcal{C}} \leftarrow \mathcal{B} [\mathbf{x}]_{\mathcal{B}} \quad (4)$$

- The columns of ${}^P_{\mathcal{C}} \leftarrow \mathcal{B}$ are the $\mathcal{C}$ -coordinate vectors of the vectors in the basis $\mathcal{B}$. That is,

$${}^P_{\mathcal{C}} \leftarrow \mathcal{B} = [[\mathbf{b}_1]_{\mathcal{C}} \quad [\mathbf{b}_2]_{\mathcal{C}} \quad \dots \quad [\mathbf{b}_n]_{\mathcal{C}}] \quad (5)$$

Change of Basis

- The matrix ${}_{\mathcal{C}} \overset{P}{\leftarrow} {}_{\mathcal{B}}$ in Theorem 15 is called the **change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** . Multiplication by ${}_{\mathcal{C}} \overset{P}{\leftarrow} {}_{\mathcal{B}}$ converts $\mathcal{B}$ -coordinates into $\mathcal{C}$ -coordinates.
- Figure 2 below illustrates the change-of-coordinates equation (4).

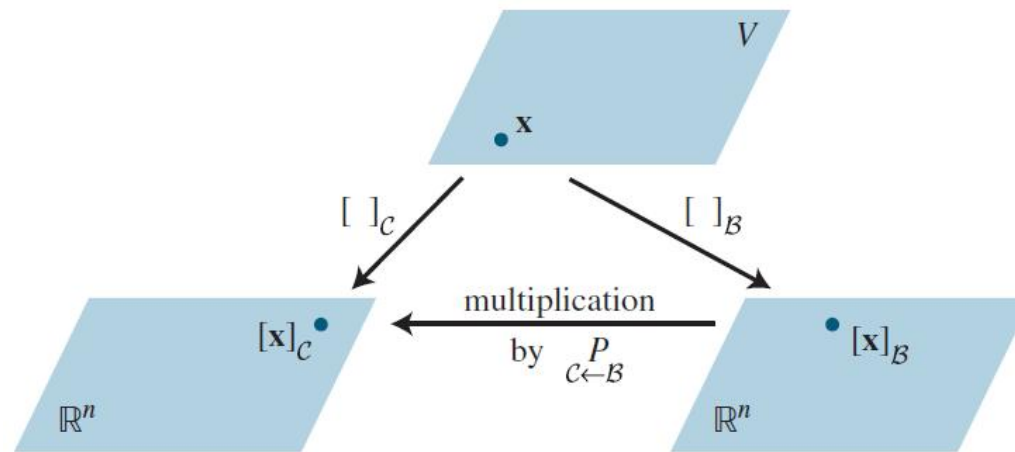


FIGURE 2 Two coordinate systems for V .

Change of Basis

- The columns of ${}^P_C \leftarrow \mathbb{B}$ are linearly independent because they are the coordinate vectors of the linearly independent set $\mathcal{B}$.
- Since ${}^P_C \leftarrow \mathbb{B}$ is square, it must be invertible, by the Invertible Matrix Theorem. Left-multiplying both sides of equation (4) by $\left({}^P_C \leftarrow \mathbb{B}\right)^{-1}$ yields

$$\left({}^P_C \leftarrow \mathbb{B}\right)^{-1} [\mathbf{x}]_C = [\mathbf{x}]_{\mathbb{B}}$$

Change of Basis

- Thus $\left(\mathcal{C} \stackrel{P}{\leftarrow} \mathcal{B} \right)^{-1}$ is the matrix that converts

$\mathcal{C}$ -coordinates into $\mathcal{B}$ -coordinates. That is, (6)

$$\left(\mathcal{C} \stackrel{P}{\leftarrow} \mathcal{B} \right)^{-1} = \mathcal{B} \stackrel{P}{\leftarrow} \mathcal{C}$$

Change of Basis in $\mathbb{R}^n$

- If $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{E}$ is the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ in $\mathbb{R}^n$, then $[\mathbf{b}_1]_{\mathcal{E}} = \mathbf{b}_1$, and likewise for the other vectors in $\mathcal{B}$. In this case, $\mathcal{E} \xleftarrow{P} \mathcal{B}$ is the same as the change-of-coordinates matrix $P_{\mathcal{B}}$ introduced in Section 4.4, namely,

$$P_{\mathcal{B}} = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \dots \quad \mathbf{b}_n]$$

- To change coordinates between two nonstandard bases in $\mathbb{R}^n$, we need Theorem 15. The theorem shows that to solve the change-of-basis problem, we need the coordinate vectors of the old basis relative to the new basis.

Change of Basis in $\mathbb{R}^n$

- **Example:** Let $\mathbf{b}_1 = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$, $\mathbf{c}_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$ and consider the bases for $\mathbb{R}^n$ given by $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2\}$. Find the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$
- **Solution** The matrix ${}_{\mathcal{C}}P_{\mathcal{B}}$ involves the $\mathcal{C}$ coordinate vectors of $\mathbf{b}_1$ and $\mathbf{b}_2$. Let $[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $[\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$. Then, by definition,

$$[\mathbf{c}_1 \quad \mathbf{c}_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{b}_1 \quad \text{and} \quad [\mathbf{c}_1 \quad \mathbf{c}_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{b}_2$$

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- To solve both systems simultaneously, augment the coefficient matrix with $\mathbf{b}_1$ and $\mathbf{b}_2$, and row reduce:

$$[\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{b}_1 \quad \mathbf{b}_2] = \begin{bmatrix} 1 & 3 & -9 & -5 \\ -4 & -5 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 6_{(7)} & 4 \\ 0 & 1 & -5 & -3 \end{bmatrix}$$

- Thus

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 6 \\ -5 \end{bmatrix} \text{ and } [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

- The desired change-of-coordinates matrix is therefore

$${}_{\mathcal{C}} \overset{P}{\leftarrow} \mathbb{B} = [[\mathbf{b}_1]_{\mathcal{C}} \quad [\mathbf{b}_2]_{\mathcal{C}}] = \begin{bmatrix} 6 & 4 \\ -5 & -3 \end{bmatrix}$$